

Anh Dung CONG

0373-716-958 | dungca1512@gmail.com | Portfolio | Blog | LinkedIn | GitHub

SUMMARY

AI/MLOps engineer with 3+ years building and operating production AI systems end-to-end: from Terraform/Ansible provisioning and Kubernetes (GKE + bare-metal kubeadm) through GitOps CI/CD to GPU model serving. Shipped speech (ASR, pronunciation/tone scoring), embedding, and agentic RAG services on GCP and AWS for users in VN, JP, KR and CN. Cost-conscious by default – GPU right-sizing, CPU-first inference paths, billing tripwires, and self-hosted alternatives to paid APIs.

EDUCATION

VNU University of Science (HUS)
Computer and Information Science
• Good academic achievement

Hanoi, Vietnam
Mar 2019 – Jun 2025

EXPERIENCE

AI/ML Systems Architect
eUp Group

Jan 2025 – Present
Hanoi, Vietnam

AI Infrastructure & MLOps ownership

Owned end-to-end AI infrastructure across the stack: provisioning (Terraform, Ansible) → containerization (Docker, Docker Compose, Docker Swarm) → orchestration (Kubernetes/GKE, Helm) → GitOps CI/CD (GitHub Actions, GitLab CI, Jenkins, ArgoCD) → GPU model serving.

Deployed and operated production ML services (ASR with faster-whisper/CTranslate2, TTS, Pronunciation Scoring with wav2vec2, Qwen3 embedding) on GKE and DigitalOcean for users in VN/JP/KR.

Migrated infrastructure from Docker Swarm to GKE; managed multi-cloud across GCP, DigitalOcean, OCI, and Cloudflare (named Tunnel, no-sudo, SSE buffering workaround).

Built a bare-metal homelab Kubernetes cluster from scratch (kubeadm) with Harbor private registry, ArgoCD (App-of-Apps), and kube-prometheus-stack (Prometheus/Grafana/Alertmanager).

Standardized release engineering across repos: Jenkins pipelines gating production on signed tags off `main` (dev builds from `develop`), pushing to GitLab Container Registry and rolling out via `docker service update` on Swarm; GitLab CI running `pytest` plus a Docker image-hygiene job (DinD build + runtime import smoke test) on every merge request.

Cost engineering: created a multi-criteria GPU sizing framework (price/region/SLA/latency) selecting GCP L4 over A100/H100 to avoid over-provisioning; performed RCA on a 3x ASR latency gap (shared vCPU + memory-bandwidth bottleneck).

Multi-market Speech Scoring Platform (Japanese JLPT, Korean TOPIK, Chinese HSKK, English)

Built and operated four FastAPI + Gunicorn scoring services sharing one architecture: multi-engine STT (Kotoba-Whisper, Komixv2 faster-whisper/CT2, SenseVoice ONNX, ReazonSpeech) with fallback, Praat/`parse1mouth` acoustic analysis under a concurrency semaphore, and parallel scoring pipelines.

Implemented language-specific scoring: Needleman–Wunsch text alignment at char/mora/bigram level and pitch-accent scoring (JA, SudachiPy/McCab/pykakasi), Mandarin tone extraction and classification (ZH), phoneme-level GOP via wav2vec2 CTC forced alignment with `espeak-ng` G2P plus stress and ending-sound detection (EN).

Kept scores deterministic (Praat/alignment-derived) and used GPT-4o-mini only for word-level feedback, preserving explainability while adding coaching value.

AI Content & Translation Systems

Hey Translate: multi-LLM translation service fanning out to Claude, Gemini, GPT, and Qwen in parallel, with strict row-alignment validation (segment-ID in/out check, no silent merge/drop) and an LLM-as-judge stage scoring adequacy/fluency/terminology to pick the best rendition.

Ultimate Lesson: Japanese lesson-generation pipeline on Gemini 2.5 Flash with a fine-tuned Gemini Flash Lite vocab extractor served from Vertex AI, quality-check + auto-retry loop, Redis caching, and a process-wide multi-key rotation pool with per-key 429 cooldown to multiply effective API quota.

Video subtitle pipeline: `yt-dlp` → Whisper ASR → GPT-4o-mini batch translation and JLPT vocabulary extraction, exposed as an async FastAPI job API with polling.

Mazii Dictionary: researched and implemented a recommendation system to personalize content on Mazii's community page.

AI Engineer

AMELA Technology

Nov 2024 – Jan 2025

Hanoi, Vietnam

Research on OCR problem for Japanese handwriting.

Research on RAG problem for chatbot.

Research on the problem of automatic cosmetic eyeliner using AI.

AI Engineer Intern

FPT Smart Cloud

Mar 2023 – Nov 2024

Hanoi, Vietnam

FPT AI Enhance for Home Credit, FE Credit, MB Bank

Participated in the development and deployment of the FPT AI Enhance project.

Handled big data processing, and trained and evaluated models for the NLP task of automatic call center scoring.

FPT AI Enhance for FPT Long Chau, FPT Shop

Built and deployed the FPT AI Enhance pipeline.

Managed big data processing, model training and evaluation, and automation of processes and data integration using n8n.

Guardrails for LLMs

Developed and tested the Guardrails project to improve the safety and control of large language models (LLMs).

Virtual Assistant Chatbot for Various Organizations

Contributed to developing and deploying virtual assistant chatbot projects for organizations such as the State Securities Commission of Vietnam, Home Credit, and FE Credit.

PROJECTS

AI Gateway

2025

Personal Project

Java 21, Spring Boot WebFlux, FastAPI, Terraform, GKE, Redis

Built a production-grade AI Gateway combining Java (Spring Boot WebFlux) gateway and Python (FastAPI) worker for unified AI request management.

Implemented multi-provider routing (OpenAI, Gemini, Claude, local LLM) with fallback, retry logic, circuit breaker, and rate limiting (Bucket4j).

Developed a Python worker supporting local LLM inference, embeddings, and a RAG pipeline with document indexing and search.

Integrated Redis caching, Prometheus monitoring, and deployed via Docker Compose locally and Kustomize on GKE.

Replaced a manual `gccloud` runbook with Terraform (API enablement, GKE Autopilot cluster, Artifact Registry, global static IP kept outside the cluster lifecycle) so the whole environment rebuilds with `apply` and drops to near-\$0 with `destroy`.

Research Agent

2025

Personal Project

Python, LangChain, LangGraph, Gemini API, MCP, A2A Protocol

Built an AI-powered research assistant for automated literature research, paper discovery, and report generation.

Designed a multi-agent architecture (Search, Paper, Synthesis agents) communicating via A2A Protocol with MCP Gateway integration.

Implemented multi-source search (Web + ArXiv), iterative research expansion, paper summarization, and structured Markdown report generation with citations.

Exposed as an MCP server for integration with Claude Desktop, Cursor, and other MCP clients.

Whisper Japanese ASR Finetune

2025

Personal Project

Python, PyTorch, HuggingFace, LoRA, Kaggle, GitHub Actions

Fine-tuned OpenAI Whisper Tiny for Japanese speech recognition using LoRA on ReazonSpeech dataset.

Built full MLOps pipeline with CI/CT/CD using GitHub Actions, Kaggle for training, and HuggingFace Hub for model hosting.

Implemented quality gate for model promotion and exported to faster-whisper (INT8) for efficient inference.

Pronunciation Scoring API

2025

eUp Group

Python, FastAPI, wav2vec2, GPU (GCP L4 / RTX 4000 Ada)

Designed a FastAPI service using wav2vec2 CTC forced alignment with a G2P caching layer and GPU micro-batching, targeting sub-3s latency for 20 concurrent users.

Sized the GPU workload (wav2vec2 315M params, FP16) to fit L4 / RTX 4000 Ada, avoiding wasteful A100/H100 provisioning.

Internal Embedding Service

2025

eUp Group

Qwen3-Embedding-4B, FastAPI, Docker Compose, RTX 4080

Deployed a self-hosted OpenAI-compatible `/v1/embeddings` service (Qwen3-Embedding-4B on RTX 4080) to fully replace the OpenAI Embedding API for internal workloads.

Built a RAG pipeline for automated code review: harvest knowledge → embed → LanceDB → inject into Qodo PR Agent, with an exit-code gate in GitLab CI that fails the pipeline on vector-store corruption.

Homelab Kubernetes & GitOps Platform

2025 – 2026

Personal Project

kubeadm, ArgoCD, Prometheus, Grafana, Loki, MetalLB, Helm

Bootstrapped a 3-node Kubernetes v1.31 cluster by hand with `kubeadm` (control-plane, etcd, kubelet, kube-proxy) rather than a managed distro, to work through every abstraction layer managed K8s hides.

Assembled the full platform: Flannel CNI, MetalLB (L2) for bare-metal LoadBalancer, ingress-nginx, local-path storage, and a self-hosted Docker registry to dodge Docker Hub rate limits.

Ran GitOps with ArgoCD App-of-Apps (multi-source Applications, prune + selfHeal) so deploys are `git push` only – no SSH, no manual `kubectl apply`.

Set up the observability stack (kube-prometheus-stack + Loki/Promtail) and documented ~2,400 lines of engineering notes, each lesson tied to a real cluster failure.

NewsPulse – Polyglot Streaming & Reco Engine

2025

Personal Project

Scala 3, Kafka, Spark Streaming, Delta Lake, Elasticsearch, Spring Boot

Designed a 7-module polyglot pipeline: Scala crawler → Kafka → Spark Streaming ETL → Python embedding service → Delta Lake, with topic clustering, real-time trending detection, Elasticsearch indexing, and a Spring Boot REST API.

Ran the whole infrastructure (Kafka, Zookeeper, Spark master/worker, Elasticsearch, Kibana, Redis) reproducibly via Docker Compose, with GitHub Actions CI and Qodana static analysis.

Cost-aware NLP + Speech Platform on GKE

2025

Personal Project

Terraform, GKE, FastAPI, SentenceTransformers, Whisper, Prometheus

Built a CPU-first inference platform (semantic search, duplicate detection, text classification, transcription) with pluggable backends, so the API, K8s manifests, and metrics flow can be developed without downloading model weights.

Provisioned GKE with a Terraform module (Workload Identity, preemptible node pool autoscaling 0→2) and shipped two container variants (lightweight vs. full-model) to keep the default deployment cheap.

Grounded RAG Advisory Assistant

2026

Client Project (under NDA)

Google ADK, FastAPI, AWS (EC2, CloudWatch), vLLM, SwiftUI

Sole engineer on a Vietnamese multi-turn advisory assistant: intent analysis, clarifying follow-up questions, ranked recommendations with explicit trade-offs, policy Q&A, and image-based product recognition.

Chose a hybrid agent architecture – the agent framework owns the runtime and session memory while orchestration stays in plain Python – after a spike showed small models hallucinate when left to self-route via tool-calling; the result is deterministic, explainable routing.

Made anti-hallucination structural rather than prompt-based: retrieval (SQLite FTS5/BM25), ranking, and every numeric value are code-owned, with a guardrail pass verifying each figure in the reply against source data, per-answer source references for click-to-verify, and PII masking in logs.

Deployed to production on **AWS EC2** behind a path-filtered GitHub Actions pipeline (`rsync` → container rebuild → health-check gate), concurrency-guarded so deploys never interleave.

Built the full observability stack on **CloudWatch** with no third-party tooling: custom metrics pushed by an on-instance `systemd` timer using an IAM instance role (no credentials on the public host), a dashboard covering cost-by-service, burstable-CPU credit burn, 5xx, health and LLM fallback rate, plus SNS-backed alarms – including a cross-region billing tripwire so cost alerts fire the moment credit runs out rather than after the bill lands.

Wrote an idempotent teardown tool (`check` / `snapshot` / `destroy` / `verify` / `restore`, with a dry-run mode) that snapshots block storage before terminating, driving idle cost to zero while keeping the stack reproducible.

Kept the LLM layer provider-agnostic (Anthropic / Gemini / self-hosted, switched by `env var`) with automatic cloud→local fallback and a rule-based offline path, plus vLLM serving scripts fitting 7B text + vision models on a single 24GB GPU via AWQ quantization and split VRAM budgets.

Shipped a native SwiftUI iOS client against the same API contract, and an evaluation harness measuring hallucination rate and follow-up quality across multi-turn scenarios alongside an automated backend test suite.

TECHNICAL SKILLS

Programming: Python, Java (17/21), Scala 3, TypeScript/Node.js, Swift (SwiftUI), Bash, SQL

GCP: GKE (Autopilot + Standard, preemptible node pools, Workload Identity), Compute Engine, Artifact Registry, Filestore, Cloud Load Balancing / static IP, IAM, Vertex AI (fine-tuned model serving) – provisioned with Terraform

AWS: EC2 & EBS (snapshot/restore lifecycle), CloudWatch (custom metrics, dashboards, alarms), SNS, IAM instance roles, billing alarms, AWS CLI automation; also S3, Lambda, Athena

Other Cloud: DigitalOcean, Oracle Cloud (OCI), Cloudflare (Pages, named Tunnel), vast.ai GPU rental

Containers & Orchestration: Docker, Docker Compose, Docker Swarm, Kubernetes (GKE + bare-metal kubeadm), Helm, ingress-nginx, MetalLB, Flannel, Harbor / self-hosted registry

IAc & CI/CD: Terraform, Ansible, GitHub Actions, GitLab CI (DinD, MR pipelines), Jenkins (tag-gated release), ArgoCD GitOps (App-of-Apps), Qodana

Observability: Prometheus, Grafana, Alertmanager, Loki, Promtail, kube-prometheus-stack

MLOps: HuggingFace Hub, LoRA/PEFT fine-tuning, CTranslate2 & ONNX quantization (INT8), CI/CT/CD training pipelines, model-promotion quality gates, GPU sizing & cost modeling, Kaggle

Speech & NLP: faster-whisper/CTranslate2, Kotoba-Whisper, SenseVoice ONNX, ReazonSpeech, wav2vec2 CTC forced alignment & GOP, Praat/parselmouth, espeak-ng G2P, SudachiPy/McCab/pykakasi, librosa, ffmpeg

LLM & Agents: LangChain, LangGraph, Google ADK, MCP, A2A Protocol, RAG (BM25/FTS5 + vector), LLM-as-judge, grounding guardrails, multi-provider routing with fallback (OpenAI/Gemini/Claude/Qwen/local), vLLM (AWQ quantization, VRAM budgeting), Ollama, llama.cpp

Frameworks: PyTorch, TensorFlow, Transformers, FastAPI, Spring Boot (WebFlux), Next.js

Data & Storage: Apache Kafka, Spark (Streaming), Delta Lake, Elasticsearch, Redis, PostgreSQL, MySQL, MongoDB, SQLite, LanceDB

Tools: Git, Linux, n8n, Makefile, pytest, Ruff

English: B1, able to communicate and read technical documents

Soft Skills: Communication, Presentation, Teamwork; able to understand and implement scientific papers

CERTIFICATIONS

Mathematics for Machine Learning and Data Science Specialization (2023)

Machine Learning Specialization (2023)

AWS Certified Solutions Architect – Associate (SAA-C03) – in progress

HONORS & AWARDS

Published 3 Whisper-based Japanese ASR models on HuggingFace Hub (dungca); LoRA variant reached 40+ downloads.

AI Gateway project submitted to the Anthropic “Claude for Open Source” program.

Author of a Vietnamese engineering blog on ML fundamentals and Kubernetes/GitOps practice (Next.js, GitHub Actions, GitHub Pages).

Second Prize in Scientific Research Project (Faculty level) – May 2022

Third Prize in Scientific Research Project (Faculty level) – May 2024

REFERENCES

Tran Trung Hieu – Senior AI Engineer, NLP team member, FPT Smart Cloud – 0945-352-916 – hieutt125@fpt.com

Nguyen Duc Quang – Senior Developer, Team leader, AMELA Technology – quang.nguyen@amela.vn